

ADRIAN E. BAYER

 [adrianbayer.github.io](https://github.com/adrianbayer)  abayer@mit.edu  0000-0002-3568-3900

 Boston, USA

ACADEMIC APPOINTMENTS

Massachusetts Institute of Technology, USA IAIFI Fellow	2026–
Center for Astrophysics Harvard & Smithsonian, USA IAIFI Fellow	2026–
Perimeter Institute, Canada Fellow	2026–
Flatiron Institute, Simons Foundation, USA Flatiron Research Fellow	2023–2026
Princeton University, USA Postdoctoral Researcher	2023–2026
The University of Tokyo, Japan Visiting Researcher (3 months)	2022
Massachusetts Institute of Technology, USA Undergraduate Researcher (2 months)	2016 Adviser: Lindley Winslow
Imperial College London, UK Undergraduate Researcher (3 months)	2015 Adviser: Henrique Araújo

EDUCATION

University of California, Berkeley, USA Ph.D. Physics	2018–2023 Thesis adviser: Uroš Seljak
University of Cambridge, UK Master of Advanced Study, Mathematics	2017–2018
Imperial College London, UK MSci Physics with Theoretical Physics Graduated top of the cohort (approx. 250 students)	2013–2017 Thesis adviser: Fay Dowker

HONORS AND AWARDS

Blavatnik Regional Award for Young Scientists in Physical Sciences and Engineering (\$10,000) Blavatnik Family Foundation and The New York Academy of Sciences	2026
Outstanding Graduate Student Instructor Award , UC Berkeley	2022
Berkeley Distinguished Graduate Fellows Video Prize (\$1,000 grant), UC Berkeley	2019
The Berkeley Fellowship , UC Berkeley	2018
Abdus Salam Undergraduate Prize , Imperial College London	2017
Governors' MSci Prize in Physics , Imperial College London	2017
Ken Allen Prize , Imperial College London	2016
Winton Capital Prize for Outstanding Performance in Second Year Physics, Imperial College London	2015
EPSRC Summer Vacation Bursary (£2,200 grant), Engineering and Physical Sciences Research Council	2015

COLLABORATIONS

Simons Observatory

Leader of Foreground / Sky Modeling Working Group

Leader of AI Initiative

Member of Digital Assets & AI Governance Committee

DESI

LSST DESC

Learning the Universe

ACADEMIC SERVICE

CMB Summer School (SOC and LOC), Flatiron Institute 2026

UrošFest – From Pen and Paper to Petaflops: Cosmology’s 30-Year Transformation (SOC) 2026

MOCK NYC (SOC), New York 2026

Astrophysics x ML Meeting (Organizer), Flatiron Institute 2025–2026

Dark Cosmos Seminar (Organizer), Princeton University 2025

Cosmology Lunch Meeting (Organizer), Princeton University 2024–2026

Cosmology and ML Meeting (Organizer), Simons Foundation 2024–2026

Numerical Cosmology and Artificial Intelligence session (Convener), COSMO’24, Japan 2024

Debating the potential of machine learning for astronomical surveys #2 (Organizer), IAP/CCA 2023

TEACHING

Machine Learning and Simulations

Lecturer, CMB Summer School, Flatiron Institute, USA 2026

Statistical Modeling and Introduction to Machine Learning

Lecturer, AstroAI Asian Network Summer School, Seoul, Korea 2025

Astronomy

Lecturer, East Jersey State Prison, Rahway, USA 2024

Bayesian Data Analysis and Machine Learning for Physical Sciences

Graduate Student Instructor, UC Berkeley, USA 2021

Python for Physics

Teaching Assistant, Imperial College London, UK 2016

MENTORING

Miranda McCarthy (Masters Student), City University of New York (2025–2026)

Career mentor

Carmen Émbil Villagr  (Graduate Student), University of Cambridge (2025–)

Project: Nonlinear information in the tSZ

Liam Parker (Graduate Student), UC Berkeley (2024–)

Project: Field-Level Inference using CNNs (co-supervised with Uro  Seljak)

Paper: [arXiv: 2504.01092](https://arxiv.org/abs/2504.01092) ([astro-ph.CO](https://arxiv.org/abs/2504.01092))

Shaunak Padhyegurjar (Undergraduate Student), IISER Bhopal, India (2024–)

Project: Primordial non-Gaussianity with Voids

Veena Krishnaraj (Undergraduate Student), Princeton University (2024–2025)

Project: Beyond the Standard Model with Machine Learning

Paper: [arXiv: 2510.19168](#) ([astro-ph.CO](#))

James Robinson (Undergraduate Student), Princeton University (2024–)

Project: Fast Algorithms for Computing the Kinetic Sunyaev-Zeldovich Effect

Arnab Lahiry (Undergraduate Student), IISER Tirupati, India (2024–2025)

Project: Interpreting the Information in the Cosmic Web (co-supervised with Francisco Villaescusa-Navarro)

Paper: [arXiv: 2504.17839](#) ([astro-ph.CO](#))

Chenze Dong (Graduate Student), University of Tokyo (2024–)

Project: Field-Level Inference for Galaxies (co-supervised with Ben Horowitz and KG Lee)

Paper: [arXiv: 2604.26823](#) ([astro-ph.CO](#))

Akira Tokiwa (Graduate Student), University of Tokyo (2023–2025)

Project: Impact of Box Size for Weak Lensing (co-supervised with Jia Liu and Masahiro Takada)

Paper: [arXiv: 2511.20423](#) ([astro-ph.CO](#))

James Sunseri (Graduate Student), Princeton University (2023–2025)

Project: Information Content of the Cosmic Web (co-supervised with Jia Liu)

Paper: [arXiv: 2503.11778](#) ([astro-ph.CO](#))

Yici Zhong (Graduate Student), University of Tokyo (2022–2024)

Project: HalfDome Cosmological Simulations for Stage IV Surveys (co-supervised with Jia Liu)

Paper: [arXiv: 2407.17462](#) ([astro-ph.CO](#))

Malika Golshan (Undergraduate Student), UC Berkeley (2022–2024)

Project: Interpreting what AI learns about neutrino physics in cosmological simulations

Paper: [arXiv: 2410.00914](#) ([astro-ph.CO](#))

Jakob Robnik (Graduate Student), UC Berkeley (2021–2024)

Project: Data-Driven Noise Models and Look-Elsewhere Effect (co-supervised with Uroš Seljak)

Paper: [arXiv: 2407.17565](#) ([astro-ph.GA](#))

REFEREEING

Journals: Astronomy & Astrophysics (A&A), The Astrophysical Journal (ApJ), Astrophysical Journal Letters (ApJL), International Conference on Machine Learning (ICML), Journal of Cosmology and Astroparticle Physics (JCAP), Monthly Notices of the Royal Astronomical Society (MNRAS), Neural Information Processing Systems (NeurIPS), Physical Review D (PRD), Physical Review Letters (PRL), Physical Review X (PRX)

Grants: ERC, IVADO

OUTREACH

NASA Moon Tree Launch Party (Invited Scientist), Madison Square Park Conservancy, New York 2026

Concrete Cosmos (Short Film Co-director)

Labocine–Simons Foundation Symbiosis program, premiered at DCTV, New York 2026

Astronomy on Tap, Jersey Shore (Organizer) 2025

Astronomy on Tap, Trenton (Host and Organizer) 2023–2025

City College of New York STEM Career Fair (Presenter) 2024

Flatiron-Nomad Partnership and Simons Foundation Public Eclipse Party (Media Representative) 2024

Berkeley Compass (Mentor) 2022

Adopt-a-Physicist (Mentor) 2020

OUTREACH TALKS

Reflections on the Moon: <i>a conversation with Tracy Scott and Denise Hill</i> Simons Foundation, New York NY, USA	December 2025
Our Learning Universe: AI Meets the Cosmos Flatiron Institute, New York NY, USA	October 2025
Neutrino Mass from Cosmology: Measuring the Mass of a Needle in a Haystack New Jersey State Museum, Planetarium, Trenton NJ, USA	August 2024
Neutrino Mass from Cosmology: Measuring the Mass of a Needle in a Haystack Astronomy on Tap, Trenton NJ, USA	November 2023

SELECTED ACADEMIC TALKS

Space Telescope Science Institute, Baltimore, MD, USA Surveying the Universe in 4D Workshop “Optimally Surveying the Universe in 4D: Machine Learning and Field-Level Inference”	August 2026
Universidade da Coruña, A Coruña, Spain New Frontiers in Cosmology: Cosmology under Totality “Field-level inference for multi-probe cosmology” & Panelist for “AI in Cosmology Discussion”	August 2026
University of Toronto, Toronto, Canada Simons Observatory F2F Meeting “Machine Learning and Simulations”	July 2026
University of Virginia, Charlottesville, VA, USA 2nd Annual NSF–Simons CosmicAI Science Conference “Simulation-based inference for multi-probe cosmology”	July 2026
Perimeter Institute for Theoretical Physics, Waterloo, Canada Advancing Field-Level and Simulation-Based Inference for Cosmology Workshop “FLISBI for LSSCMB” & panelist for AI discussion	June 2026
UrošFest, San Francisco, CA, USA From Pen and Paper to Petaflops: Cosmology’s 30-Year Transformation “Reconstructing Grad School with Uros-Level Inference” & chair and panelist for AI Discussion	May 2026
60th Rencontres de Moriond, La Thuile, Italy Cosmology Conference “Extracting optimal and robust information from cosmological surveys with simulation-based inference”	March 2026
Kavli Institute for Theoretical Physics, UC Santa Barbara, CA, USA Exploring New Boundaries in Cosmology and Astrophysics: CMB and LSS Workshop “Simulations for ML Applications: Prospects and Challenges for Multi-Probe Simulation-Based Inference”	February 2026
University of Texas, Austin, TX, USA Weinberg Seminar “Unifying the Sky: Field-Level Inference for Multi-Probe Cosmology”	February 2026
University of Texas, Austin, TX, USA Physics Colloquium “Why a tiny neutrino particle inspired me to simulate and reconstruct the entire Universe”	February 2026
Kavli Institute for Particle Astrophysics & Cosmology, Stanford University, CA, USA KIPAC Seminar “Field-Level Inference in the Multimodal Cosmos: Scaling Scientific Discovery Across Fields with AI”	February 2026

Kavli Institute for Theoretical Physics, UC Santa Barbara, CA, USA New Synergies in Multi-Probe Cosmology Conference “A Review of Machine Learning and AI for Cosmology and Astrophysics”	February 2026
Institute of Astronomy and Astrophysics, Academia Sinica, Taipei, Taiwan Reconstructing the Universe Workshop “Field-Level BAO Reconstruction and beyond”	December 2025
Center for Computational Astrophysics, Flatiron Institute, New York NY, USA Cosmology X Data Science Meeting “Why a tiny neutrino particle inspired me to simulate and reconstruct the entire Universe”	November 2025
Rutgers University, New Brunswick NJ, USA New High Energy Theory Center Seminar “Why a tiny neutrino particle inspired me to simulate and reconstruct the entire Universe”	November 2025
University of Waterloo, Waterloo, Canada WCA Astroseminar “What’s the Likelihood? Field-Level Cosmology x Detecting Signals in Large and Noisy Places”	October 2025
Perimeter Institute, Waterloo, Canada Cosmology Meeting “Why a tiny neutrino particle inspired me to simulate and reconstruct the entire Universe”	October 2025
ETH Zürich, Zürich, Switzerland CosmoClub Seminar “Why a tiny neutrino particle inspired me to simulate and reconstruct the entire Universe”	September 2025
Erwin Schrödinger International Institute, Vienna, Austria Putting the Cosmic Large-scale Structure on the Map: Theory Meets Numerics “Field-Level BAO Reconstruction and Beyond”	September 2025
Max Planck Institute for Astrophysics, Munich, Germany Cosmology Meeting “Field-Level BAO Reconstruction and Beyond”	August 2025
University of Cambridge (Kavli Institute for Cosmology), Cambridge, UK CMB/LSS Meeting “Field-Level BAO Reconstruction and Beyond”	August 2025
Centro de Ciencias de Benasque Pedro Pascual, Benasque, Spain Understanding Cosmological Observations Workshop “Field-Level BAO Reconstruction”	July 2025
University of Manchester, UK Simons Observatory Collaboration Meeting “Sky Modeling”	July 2025
Sexten Center for Astrophysics, Sexten, Italy New Strategies For Extracting Cosmology From Future Galaxy Surveys Workshop – 3rd Edition “Field-Level BAO Reconstruction”	July 2025
Center for Computational Astrophysics, Flatiron Institute, New York NY, USA Exoplanet Meeting “Detecting astronomical signals in large and noisy spaces”	June 2025
Center for Computational Astrophysics, Flatiron Institute, New York NY, USA Cosmology X Data Science Meeting “What’s the Likelihood? Field-Level Cosmology x Detecting Signals in Large and Noisy Places”	May 2025

Center for Astrophysics, Harvard University, Cambridge MA, USA AstroAI Seminar “Why a tiny neutrino particle inspired me to reconstruct the initial conditions of the Universe”	April 2025
Center for Astrophysics, Harvard University, Cambridge MA, USA ITC Luncheon Talk “Field-Level Reconstruction of the Cosmological Initial Conditions”	April 2025
Massachusetts Institute of Technology, Cambridge MA, USA TESS Science Talk “Detecting astronomical signals in large and noisy spaces”	April 2025
University of Pennsylvania, Philadelphia PA, USA Astro Seminar “Why a tiny neutrino particle inspired me to simulate and reconstruct the entire Universe”	April 2025
Tokyo University, Tokyo, Japan Astronomy Seminar “Towards an Optimal Cosmological Detection of Neutrino Mass with Field-Level Inference and Joint Analyses”	November 2024
Kyoto University, Kyoto, Japan COSMO’24 “Introduction to Numerical Cosmology and Artificial Intelligence”	October 2024
University of Chicago, Chicago IL, USA Simons Observatory Collaboration Meeting “CMB x LSS with the HalfDome Simulations”	July 2024
Mediterranean Institute for Life Sciences, Split, Croatia Cosmology in the Adriatic – From PT to AI “Cosmology in the Adriatic with Adrian: from field-level inference to joint analyses”	July 2024
Università degli Studi di Catania - Dipartimento di Fisica e Astronomia, Catania, Italy International Conference on Machine Learning for Astrophysics – 2nd Edition “Extracting optimal information from cosmological surveys with field-level inference and joint analyses”	July 2024
Sexten Center for Astrophysics, Sexten, Italy New Strategies For Extracting Cosmology From Future Galaxy Surveys Workshop – 2nd Edition “The HalfDome CMB x LSS Simulations”	July 2024
Aspen Center for Physics, Aspen CO, USA Fundamental Physics in the Era of Big Data and Machine Learning “Physics-based sampling”	June 2024
Grand Arsenal, Chania, Greece COSMO21: Statistical Challenges in 21st Century Cosmology “Towards an Optimal Cosmological Detection of Neutrino Mass with Joint Analyses and Field-Level Inference”	May 2024
Stanford University, Stanford CA, USA Cosmology Seminar “Towards an Optimal Cosmological Detection of Neutrino Mass with Field-Level Inference”	April 2024
Yale University, New Haven CT, USA Cosmology Seminar “Towards an Optimal Cosmological Detection of Neutrino Mass with Field-Level Inference”	April 2024
Center for Computational Astrophysics, Flatiron Institute, New York NY, USA Cosmology X Data Science Meeting “The HalfDome CMB x LSS Simulations”	March 2024

The Center for Cosmology and Particle Physics, New York University, NY, USA Astrophysics and Relativity Seminar “Towards an Optimal Cosmological Detection of Neutrino Mass with Field-Level Inference”	March 2024
Center for Data Driven Discovery (CD3), Kavli IPMU, University of Tokyo, Japan The CD3 x Simons Foundation workshop: AI-driven discovery in physics and astrophysics “Cosmological Field-Level Inference with Microcanonical Langevin Monte Carlo”	January 2024
High Energy Accelerator Research Organization (KEK), Tsukuba, Japan ML at HEP workshop “Extracting optimal information from upcoming cosmological surveys”	January 2024
Center for Computational Astrophysics, Flatiron Institute, New York NY, USA Debating the potential of machine learning for astronomical surveys (#2) – IAP/CCA Conference “Cosmological Field-Level Inference with Microcanonical Langevin Monte Carlo”	November 2023
Imperial College London, London, UK Seminar “Towards an Optimal Cosmological Detection of Neutrino Mass”	November 2023
Monte Verità, Ascona, Switzerland Hamers & Nails, Frontiers in Machine Learning in Cosmology, Astro & Particle Physics “Cosmological Field-Level Inference with Microcanonical Langevin Monte Carlo”	October 2023
Hawaii Convention Center, Honolulu HI, USA International Conference on Machine Learning (ICML), Workshop on Machine Learning for Astrophysics “Field-Level Inference with Microcanonical Langevin Monte Carlo”	July 2023
Sexten Center for Astrophysics, Sexten, Italy New Strategies For Extracting Cosmology From Future Galaxy Surveys Workshop “Field-Level Inference with Microcanonical Hamiltonian Monte Carlo”	July 2023
Center for Computational Astrophysics, Flatiron Institute, New York NY, USA Cosmic Connections Symposium “Field-Level Inference with Microcanonical Hamiltonian Monte Carlo”	May 2023
Stanford University, Stanford CA, USA (zoom) LSST Higher-Order Statistics Meeting “Super-Sample Covariance of Higher-Order Statistics”	January 2023
Institute for Advanced Studies, Princeton NJ, USA Cosmology Lunch “Towards Optimal Measurement of the Neutrino Mass using Large-Scale Structure”	December 2022
Center for Computational Astrophysics, Flatiron Institute, New York NY, USA Cosmology X Data Science Meeting “Towards Optimal Measurement of the Neutrino Mass using Large-Scale Structure”	December 2022
University of Pennsylvania, Philadelphia PA, USA CMB Meeting “Towards Optimal Measurement of the Neutrino Mass using Large-Scale Structure”	December 2022
Université de Montréal, Montréal, Canada Astromerique Speaker Series “Massive Neutrino Information in Large-Scale Structure and Field-Level Inference”	November 2022
Vipolže, Slovenia Berkeley Center for Cosmological Physics Summer Workshop “Massive Neutrino Information in Large-Scale Structure and Field-Level Inference”	July 2022

The University of Tokyo (Hongo Campus), Tokyo, Japan GPU Workshop “Cosmological simulations on GPU with tensorflow”	May 2022
The University of Tokyo (Kavli IPMU), Kashiwanoha, Japan APEC Seminar “Towards detecting neutrino mass using non-linear cosmic structure”	April 2022
Kyoto University (Yukawa Institute for Theoretical Physics), Kyoto, Japan Cosmology with Weak Lensing: Beyond the 2-pt Statistics “Detecting neutrino mass using nonlinear cosmic structure”	April 2022
Institut d’Astrophysique de Paris, Paris, France Journal Club – Univers “Detecting neutrino mass using nonlinear cosmic structure”	February 2022
Center for Computational Astrophysics, Flatiron Institute, New York NY, USA Learn the Universe “The Look-Elsewhere Effect”	August 2021
Pennsylvania State University, State College PA, USA Statistical Challenges in Modern Astronomy VII “The Look-Elsewhere Effect from a Unified Bayesian and Frequentist Perspective”	June 2021
University of Cambridge (Kavli Institute for Cosmology), Cambrigde, UK KICC 10th Anniversary Symposium “Look Elsewhere” (poster)	September 2019

BIBLIOGRAPHY

PRIMARY AUTHOR:

1. Parker, L., U. Seljak, **A. E. Bayer**, R. Feder, and D. Valcin. “Field-Level Baryon Acoustic Oscillation Reconstruction of the DESI DR1 Luminous Red Galaxies with Linear Field Transformer (LiFT)”. in: (Sept. 2026). arXiv: [2609.14862 \[astro-ph.CO\]](#)
2. **Bayer, A. E.**, L. Parker, D. Valcin, S.-F. Chen, C. Modi, and U. Seljak. “Field-Level Inference from Galaxies: BAO Reconstruction”. In: (Mar. 2026). arXiv: [2603.15732 \[astro-ph.CO\]](#)
3. Tokiwa, A., **A. E. Bayer**, J. Armijo, J. Liu, R. Terasawa, L. Thiele, M. Alvarez, L. Blot, and M. Takada. “Impact of Simulation Box Size for Weak Lensing: Replication and Super-Sample Effects”. In: (Nov. 2025). arXiv: [2511.20423 \[astro-ph.CO\]](#)
4. Horowitz, B. and **A. E. Bayer**. “jFoF: GPU Cluster Finding with Gradient Propagation”. In: (Oct. 2025). arXiv: [2510.26851 \[astro-ph.IM\]](#)
5. Krishnaraj, V., **A. E. Bayer**, C. K. Jespersen, and P. Melchior. “Transfer Learning Beyond the Standard Model”. In: (Oct. 2025). arXiv: [2510.19168 \[astro-ph.CO\]](#)
6. Thiele, L., **A. E. Bayer**, and N. Takeishi. “Simulation-Efficient Cosmological Inference with Multi-Fidelity SBI”. in: (July 2025). arXiv: [2507.00514 \[astro-ph.CO\]](#)
7. **Bayer, A. E.**, F. Villaescusa-Navarro, S. Sharief, R. Teyssier, L. H. Garrison, L. Perreault-Levasseur, G. L. Bryan, M. Gatti, and E. Visbal. “Field-level Comparison and Robustness Analysis of Cosmological N-body Simulations”. In: *ApJ* 989.2 (2025), p. 207. DOI: [10.3847/1538-4357/ade44e](#). arXiv: [2505.13620 \[astro-ph.CO\]](#)
8. Lahiry, A., **A. E. Bayer**, and F. Villaescusa-Navarro. “Interpreting Cosmological Information from Neural Networks in the Hydrodynamic Universe”. In: *The Astrophysical Journal* 994.1 (Apr. 2025), p. 129. DOI: [10.3847/1538-4357/ae0b5a](#). arXiv: [2504.17839 \[astro-ph.CO\]](#)

9. Parker, L., **A. E. Bayer**, and U. Seljak. “Initial conditions from galaxies: machine-learning subgrid correction to standard reconstruction”. In: *JCAP* 09 (2025), p. 039. DOI: [10.1088/1475-7516/2025/09/039](https://doi.org/10.1088/1475-7516/2025/09/039). arXiv: [2504.01092](https://arxiv.org/abs/2504.01092) [[astro-ph.CO](#)]
10. Sunseri, J., **A. E. Bayer**, and J. Liu. “Power of the cosmic web”. In: *Phys. Rev. D* 112.6 (2025), p. 063516. DOI: [10.1103/grx3-hj7w](https://doi.org/10.1103/grx3-hj7w). arXiv: [2503.11778](https://arxiv.org/abs/2503.11778) [[astro-ph.CO](#)]
11. Cuesta-Lazaro, C., **A. E. Bayer**, M. S. Albergo, S. Mishra-Sharma, C. Modi, and D. J. Eisenstein. “Joint cosmological parameter inference and initial condition reconstruction with Stochastic Interpolants”. In: *NeurIPS 2024 Workshop: Machine Learning and the Physical Sciences*. 2024. URL: https://ml4physicalsciences.github.io/2024/files/NeurIPS_ML4PS_2024_162.pdf
12. Golshan, M. and **A. E. Bayer**. “Massive ν s through the CNN lens: interpreting the field-level neutrino mass information in weak lensing”. In: *JCAP* 05 (2025), p. 024. DOI: [10.1088/1475-7516/2025/05/024](https://doi.org/10.1088/1475-7516/2025/05/024). arXiv: [2410.00914](https://arxiv.org/abs/2410.00914) [[astro-ph.CO](#)]
13. Robnik, J., **A. E. Bayer**, M. Charisi, Z. Haiman, A. Lin, and U. Seljak. “Periodicity significance testing with null-signal templates: reassessment of PTF’s SMBH binary candidates”. In: *MNRAS* 534.2 (Oct. 2024), pp. 1609–1620. DOI: [10.1093/mnras/stae2220](https://doi.org/10.1093/mnras/stae2220). arXiv: [2407.17565](https://arxiv.org/abs/2407.17565) [[astro-ph.GA](#)]
14. **Bayer, A. E.**, Y. Zhong, Z. Li, J. DeRose, Y. Feng, and J. Liu. “The HalfDome multi-survey cosmological simulations: N-body simulations”. In: *JCAP* 05 (2025), p. 016. DOI: [10.1088/1475-7516/2025/05/016](https://doi.org/10.1088/1475-7516/2025/05/016). arXiv: [2407.17462](https://arxiv.org/abs/2407.17462) [[astro-ph.CO](#)]
15. **Bayer, A. E.**, J. Liu, C. D. Kreisch, and A. Pisani. “Significance of void shape: Neutrino mass from Voronoi void halos?” In: *Phys. Rev. D* 110.6, L061305 (Sept. 2024), p. L061305. DOI: [10.1103/PhysRevD.110.L061305](https://doi.org/10.1103/PhysRevD.110.L061305). arXiv: [2405.12302](https://arxiv.org/abs/2405.12302) [[astro-ph.CO](#)]
16. **Bayer, A. E.**, U. Seljak, and C. Modi. “Field-Level Inference with Microcanonical Langevin Monte Carlo”. In: *40th International Conference on Machine Learning*. July 2023. arXiv: [2307.09504](https://arxiv.org/abs/2307.09504) [[astro-ph.CO](#)]
17. **Bayer, A. E.**, C. Modi, and S. Ferraro. “Joint velocity and density reconstruction of the Universe with nonlinear differentiable forward modeling”. In: *J. Cosmology Astropart. Phys.* 2023.6, 046 (June 2023), p. 046. DOI: [10.1088/1475-7516/2023/06/046](https://doi.org/10.1088/1475-7516/2023/06/046). arXiv: [2210.15649](https://arxiv.org/abs/2210.15649) [[astro-ph.CO](#)]
18. **Bayer, A. E.**, J. Liu, R. Terasawa, A. Barreira, Y. Zhong, and Y. Feng. “Super-sample covariance of the power spectrum, bispectrum, halos, voids, and their cross covariances”. In: *Phys. Rev. D* 108.4 (2023), p. 043521. DOI: [10.1103/PhysRevD.108.043521](https://doi.org/10.1103/PhysRevD.108.043521). arXiv: [2210.15647](https://arxiv.org/abs/2210.15647) [[astro-ph.CO](#)]
19. **Bayer, A. E.**, A. Banerjee, and U. Seljak. “Beware of fake ν ’s: The effect of massive neutrinos on the nonlinear evolution of cosmic structure”. In: *Phys. Rev. D* 105.12, 123510 (June 2022), p. 123510. DOI: [10.1103/PhysRevD.105.123510](https://doi.org/10.1103/PhysRevD.105.123510). arXiv: [2108.04215](https://arxiv.org/abs/2108.04215) [[astro-ph.CO](#)]
20. **Bayer, A. E.**, U. Seljak, and J. Robnik. “Self-calibrating the look-elsewhere effect: fast evaluation of the statistical significance using peak heights”. In: *MNRAS* 508.1 (Nov. 2021), pp. 1346–1357. DOI: [10.1093/mnras/stab2331](https://doi.org/10.1093/mnras/stab2331). arXiv: [2108.06333](https://arxiv.org/abs/2108.06333) [[astro-ph.IM](#)]
21. **Bayer, A. E.**, F. Villaescusa-Navarro, E. Massara, J. Liu, D. N. Spergel, L. Verde, B. D. Wandelt, M. Viel, and S. Ho. “Detecting Neutrino Mass by Combining Matter Clustering, Halos, and Voids”. In: *ApJ* 919.1, 24 (Sept. 2021), p. 24. DOI: [10.3847/1538-4357/ac0e91](https://doi.org/10.3847/1538-4357/ac0e91). arXiv: [2102.05049](https://arxiv.org/abs/2102.05049) [[astro-ph.CO](#)]
22. **Bayer, A. E.**, A. Banerjee, and Y. Feng. “A fast particle-mesh simulation of non-linear cosmological structure formation with massive neutrinos”. In: *J. Cosmology Astropart. Phys.* 2021.1, 016 (Jan. 2021), p. 016. DOI: [10.1088/1475-7516/2021/01/016](https://doi.org/10.1088/1475-7516/2021/01/016). arXiv: [2007.13394](https://arxiv.org/abs/2007.13394) [[astro-ph.CO](#)]
23. **Bayer, A. E.** and U. Seljak. “The look-elsewhere effect from a unified Bayesian and frequentist perspective”. In: *J. Cosmology Astropart. Phys.* 2020.10, 009 (Oct. 2020), p. 009. DOI: [10.1088/1475-7516/2020/10/009](https://doi.org/10.1088/1475-7516/2020/10/009). arXiv: [2007.13821](https://arxiv.org/abs/2007.13821) [[physics.data-an](#)]

CONTRIBUTING AUTHOR:

1. Petric, A. et al. *Surveying the Universe in 4D: Beating Cosmic Variance with Wide-Field Slitless Spectroscopy from HST, JWST, Euclid, Roman, and Beyond*. Sept. 2026. arXiv: [2609.23148 \[astro-ph.GA\]](#)
2. Liu, J. et al. “AI’s Capability in Assisting Scientific Research in Physics, Astrophysics, and Cosmology II: Project Planning and Proposal Evaluation”. In: (July 2026). arXiv: [2607.25881 \[cs.CL\]](#)
3. Hell, A. et al. “AI’s Capability in Assisting Scientific Research in Physics, Astrophysics, and Cosmology I: Literature Review”. In: (July 2026). arXiv: [2607.25672 \[astro-ph.IM\]](#)
4. Dong, C., B. Horowitz, **A. E. Bayer**, and K.-G. Lee. “MujicA: Reconstructing Initial Conditions from Incomplete Redshift Surveys with Projected Optimization”. In: (Apr. 2026). arXiv: [2604.26823 \[astro-ph.CO\]](#)
5. Duivenvoorden, A. J., K. Surrao, **A. E. Bayer**, A. E. Adler, N. Dachlythra, S. Azzoni, and J. C. Hill. “Robust CMB B-mode analysis with Needlet-ILC and simulation-based inference”. In: (Dec. 2025). arXiv: [2512.16869 \[astro-ph.CO\]](#)
6. The Simons Observatory Collaboration. “The Simons Observatory: forecasted constraints on primordial gravitational waves with the expanded array of Small Aperture Telescopes”. In: (Dec. 2025). arXiv: [2512.15833 \[astro-ph.CO\]](#)
7. Villaescusa-Navarro, F., B. Bolliet, P. Villanueva-Domingo, **A. E. Bayer**, A. Acquah, C. Amancharla, A. Barzilai-Siegal, P. Bermejo, C. Bilodeau, P. C. Ramírez, M. Cranmer, U. L. França, C. Hahn, Y.-F. Jiang, R. Jimenez, J.-Y. Lee, A. Lerario, O. Mamun, T. Meier, A. A. Ojha, P. Protopapas, S. Roy, D. N. Spergel, P. Tarancón-Álvarez, U. Tiwari, M. Viel, D. Wadekar, C. Wang, B. Y. Wang, L. Xu, Y. Yovel, S. Yue, W.-H. Zhou, Q. Zhu, J. Zou, and Í. Zubeldia. “The Denario project: Deep knowledge AI agents for scientific discovery”. In: (Oct. 2025). arXiv: [2510.26887 \[cs.AI\]](#)
8. Crespi, A., M. Bonici, A. Loureiro, J. Ruiz-Zapatero, I. Sladoljev, Z. Li, **A. Bayer**, M. Millea, and U. Seljak. “Flinch: A Differentiable Framework for Field-Level Inference of Cosmological parameters from curved sky data”. In: (Oct. 2025). arXiv: [2510.26691 \[astro-ph.CO\]](#)
9. Xu, L., M. Sarkar, A. I. Lonappan, Í. Zubeldia, P. Villanueva-Domingo, S. Casas, C. Fidler, C. Amancharla, U. Tiwari, **A. Bayer**, C. A. Ekiou, M. Cranmer, A. Dimitrov, J. Fergusson, K. Gandhi, S. Krippendorf, A. Laverick, J. Lesgourgues, A. Lewis, T. Meier, B. Sherwin, K. Surrao, F. Villaescusa-Navarro, C. Wang, X. Xu, and B. Bolliet. “Open Source Planning & Control System with Language Agents for Autonomous Scientific Discovery”. In: (2025). arXiv: [2507.07257 \[cs.AI\]](#). URL: <https://arxiv.org/abs/2507.07257>
10. Huang, N., R. Stiskalek, J.-Y. Lee, **A. E. Bayer**, C. C. Margossian, C. K. Jespersen, L. A. Perez, L. K. Saul, and F. Villaescusa-Navarro. “CosmoBench: A Multiscale, Multiview, Multitask Cosmology Benchmark for Geometric Deep Learning”. In: (2025). arXiv: [2507.03707 \[cs.LG\]](#). URL: <https://arxiv.org/abs/2507.03707>
11. Zeghal, J., D. Lanzieri, F. Lanusse, A. Boucaud, G. Louppe, E. Aubourg, **A. E. Bayer**, and The LSST Dark Energy Science Collaboration. “Simulation-Based Inference Benchmark for LSST Weak Lensing Cosmology”. In: (2024). arXiv: [2409.17975 \[astro-ph.CO\]](#). URL: <https://arxiv.org/abs/2409.17975>
12. Pandey, S., C. Modi, B. D. Wandelt, D. J. Bartlett, **A. E. Bayer**, G. L. Bryan, M. Ho, G. Lavaux, T. L. Makinen, and F. Villaescusa-Navarro. “Creating halos with autoregressive multistage networks”. In: *Phys. Rev. D* 112.10 (2025), p. 103503. DOI: [10.1103/vlm2-tm6k](#)
13. Ding, Z., C.-H. Chuang, Y. Yu, L. H. Garrison, **A. E. Bayer**, Y. Feng, C. Modi, D. J. Eisenstein, M. White, A. Variu, C. Zhao, H. Zhang, J. Meneses Rizo, D. Brooks, K. Dawson, P. Doel, E. Gaztanaga, R. Kehoe, A. Krolewski, M. Landriau, N. Palanque-Delabrouille, and C. Poppett. “The DESI N-body Simulation Project - II. Suppressing sample variance with fast simulations”. In: *MNRAS* 514.3 (Aug. 2022), pp. 3308–3328. DOI: [10.1093/mnras/stac1501](#). arXiv: [2202.06074 \[astro-ph.CO\]](#)
14. Kreisch, C. D., A. Pisani, F. Villaescusa-Navarro, D. N. Spergel, B. D. Wandelt, N. Hamaus, and **A. E. Bayer**. “The GIGANTES Data Set: Precision Cosmology from Voids in the Machine-learning Era”. In: *ApJ* 935.2, 100 (Aug. 2022), p. 100. DOI: [10.3847/1538-4357/ac7d4b](#). arXiv: [2107.02304 \[astro-ph.CO\]](#)

15. Tomás, A., H. M. Araújo, A. J. Bailey, **A. Bayer**, E. Chen, B. López Paredes, and T. J. Sumner. “Study and mitigation of spurious electron emission from cathodic wires in noble liquid time projection chambers”. In: *Astroparticle Physics* 103 (Dec. 2018), pp. 49–61. DOI: [10.1016/j.astropartphys.2018.07.001](https://doi.org/10.1016/j.astropartphys.2018.07.001). arXiv: [1801.07231](https://arxiv.org/abs/1801.07231) [[physics.ins-det](#)]

REFERENCES

Prof. Uroš Seljak

useljak@berkeley.edu

Departments of Physics and Astronomy, University of California, Berkeley

Prof. David Spergel

dspergel@simonsfoundation.org

Simons Foundation, New York City

Prof. Jia Liu

jia.liu@ipmu.jp

Kavli Institute for the Physics and Mathematics of the Universe, University of Tokyo

Dr. Francisco Villaescusa-Navarro

fvillaescusa@flatironinstitute.org

Center for Computational Astrophysics, Flatiron Institute, New York City